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v7.15 sparse chloride-occlusion extension

Lifecycle: human-ratified implementation contract for the v7.15 candidate. This document governs the bounded implementation semantics but does not self-qualify the final implementation; independent post-implementation review remains required.

Status and scope

This document is the canonical scientific/numerical implementation contract for the v7.15 sparse chloride-occlusion feature. It extends only the supported Al24Si24O96 LTA primitive-cell generator. It does not add non-Al24 frameworks, dense molten-salt packing, thermodynamic sampling, a salt-loaded empirical volume model, arbitrary-setting coordinate remapping, or joint cation-plus-cage maximum-symmetry certification.

The construction separates three concepts:

1. framework-compensating Group-I/II cations on grouped 4R/6R/8R sites;
2. additional `occlusion_counterion` cations initialized on the same physical grouped-site inventory; and
3. chloride cage guests initialized at explicitly supported alpha/beta centers.

The role split is construction/provenance metadata. It is not a claim that relaxed electrostatics remain spatially separable.

Independent charge invariants

For formal cation charge `z_i`:

```
Q_fw  = sum_i z_i N_i^fw  = 24
N_Cl  = N_Cl^alpha + N_Cl^beta
Q_occ = sum_i z_i N_i^occ = N_Cl
```

Therefore the complete initialized composition also satisfies

```
sum_i z_i (N_i^fw + N_i^occ) = 24 + N_Cl
```

but this global equality is not a substitute for either independent invariant. The legacy `alkali_counts` field remains the framework-compensating population. The automatic dry-volume interpolation consumes only this framework population. The derived `framework_cation_counts` view is likewise framework-only, while `cation_counts` (and its explicit alias `total_cation_counts`) reports the total physical cation population across both roles.

Sparse chloride domain and canonical setting

v7.15 accepts exact integer counts

```
0 <= alpha_cl <= 2
0 <= beta_cl  <= 2
```

with at most one Cl- at each supported center.

For the bound canonical Fm-3c primitive fixture `tests/real_lta_primitive.cif`:

- beta: `(0,0,0)`, `(1/2,1/2,1/2)`;
- alpha: `(1/4,1/4,1/4)`, `(3/4,3/4,3/4)`.

Occlusion is fail-closed to that bound template identity and the expected two-member alpha/beta center orbits. Shifted, rebased, re-originated, or otherwise incompatible templates are rejected until an explicit coordinate mapping is reviewed and implemented. These cage centers are AIMD initialization conventions, not equilibrium-coordinate predictions.

For a requested count of one in a family, the selected center maximizes its minimum periodic Cartesian distance to the final cation coordinates. Exact geometric ties use canonical cage-ID order. A requested count of two occupies both centers.

Cation placement and RNG isolation

Framework and occlusion-counterion roles share the same grouped physical-site capacities:

```
sum_i (N_ri^fw + N_ri^occ) <= C_r
```

Framework placement occurs first with the legacy random stream. Occlusion counterions are placed only on remaining empty requested sites with a deterministic secondary stream derived from the user seed. Consequently, for a fixed seed and fixed framework allocation, enabling occlusion does not consume or perturb the framework placement stream.

There is no v7.15 automatic occlusion rebalancing.

Collision guard

After final cation geometry is fixed, Cl cage centers are selected and every Cl is checked against the complete framework/cation/Cl structure by minimum-image periodic Cartesian distance. The default `--occlusion-min-distance` is 1.5 Angstrom. This is only a catastrophic overlap guard; it is not a preferred Cl-X bond or contact distance. Failure rejects the configuration rather than moving Cl away from the cage-center model.

Dry-volume compatibility boundary

The established v7.14 dry reference remains

```
V_ref = (1/24) sum_i z_i N_i^fw V_i^(24)
V_target = volume_factor V_ref
```

Only framework-compensating counts enter this equation. Extra counterions and Cl guests do not alter the automatic reference volume in v7.15. This is an initialization policy preserving the qualified dry model, not a scientific claim that chloride loading has zero physical volume effect.

Public interface and unsupported combinations

Both random-generation entry points expose:

- `--alpha-cl INT`, `--beta-cl INT`;
- `--occlusion-n4 SPEC`, `--occlusion-n6 SPEC`, `--occlusion-n8 SPEC`;
- `--occlusion-min-distance FLOAT` (default 1.5 Angstrom).

When occlusion is active, framework and excess allocations must be exact counts. v7.15 rejects occlusion with ratio allocation, automatic rebalancing, maximum-symmetry mode, symmetry-unique batches larger than one, and candidate-variant output.

MS0-MS7 remain cation-symmetry machinery. No certificate produced by those paths establishes maximum symmetry of a structure containing chloride guests, so nonzero occlusion is rejected before maximum-symmetry optimization or certification.

Smoothing and assembly

Gaussian-density analysis remains a cation-density heuristic. Cl is excluded from the density estimator. The random smoothing order is:

1. place framework cations;
2. place excess-role cations;
3. smooth cations;
4. select/place chloride after the final cation geometry;
5. apply the chloride collision guard;
6. assemble and serialize the complete structure.

There is one physical assembly path: framework atoms + all grouped-site cations + chloride cage guests. CIF, POSCAR/VASP, and LAMMPS writers serialize that same structure. Cl follows the existing Group-I/II cations in preferred species order. LAMMPS remains `atom_style atomic`; no force-field partial charges are introduced.

Epistemic scope of the literature

Lewis, Fischer, and Smith experimentally studied salt-occluded zeolite A as a matrix for chloride waste salt and describe chloride salt loaded into zeolite cavities, with Cs and Sr preferentially sorbed. This supports salt occlusion in LTA as a physical phenomenon, but it does not by itself establish the present cage-center coordinates as equilibrium sites.

Simpson and Gougar fitted a two-site molten-chloride/zeolite-A ion-exchange model for monovalent cations and reported distinct framework and alpha-cage selectivities. Phongikaroon and Simpson extended the two-site model to multivalent cations and distinguished ion-exchange and occlusion-site selectivities. These sources motivate a controlled separation between framework-compensating and excess/occlusion-role initialization populations; they do not establish every grouped 4R/6R/8R coordinate as an experimental molten-salt occlusion site.

Alpha-cage Cl in v7.15 is therefore a sparse initialization informed by established salt occlusion in LTA. Beta-cage Cl is deliberately sampled as a trapped/configurational **training hypothesis** and must not be described as a demonstrated equilibrium chloride site.

References

1. M. A. Lewis, D. F. Fischer, and L. J. Smith, "Salt-Occluded Zeolites as an Immobilization Matrix for Chloride Waste Salt," *J. Am. Ceram. Soc.* 76, 2826-2832 (1993). DOI: 10.1111/j.1151-2916.1993.tb04023.x.
2. M. F. Simpson and M. L. D. Gougar, "Two-Site Equilibrium Model for Ion Exchange between Monovalent Cations and Zeolite-A in a Molten Salt," *Ind. Eng. Chem. Res.* 42, 4208-4212 (2003). DOI: 10.1021/ie020896b.
3. S. Phongikaroon and M. F. Simpson, "Equilibrium model for ion exchange between multivalent cations and zeolite-A in a molten salt," *AIChE J.* 52, 1736-1743 (2006). DOI: 10.1002/aic.10784.

All existing v7.14 chemistry, crystallography, dry-volume, pymatgen, LAMMPS, symmetry, and algorithmic references remain applicable within their prior scopes.